Faculty of Computer Science

Applied Quantitative Logistics

Moscow 2025

A Hybrid GWO and GAPSO Optimized Approach for Heart-Disease Prediction By RF and XGB Classifier

By:
Amit Raj & Salman Hussain

Authors: El-Shafieyet al. **Publication:** Springer(2022)
Paper Link: httpshttps://link.springer.com/article/10.1007/s11042-022-12425-x

Introduction

- Heart disease is one of the leading causes of death worldwide.
- Early detection is crucial to improving outcomes and saving lives.
- In this final project, we built upon our intermediate work and implemented:
  - GAPSO with XGB classifier
  - Introduced GWO optimizer
  - Used two datasets: Cleveland and Statlog
  - Compared performance using RF and XGB classifiers

Objective of the Study:

- Select the most informative features using GAPSO and GWO.
- Evaluate classifiers: Random Forest (RF) and XGBoost (XGB).
- Achieve higher accuracy with optimized models.
- Compare performance on two heart disease datasets.

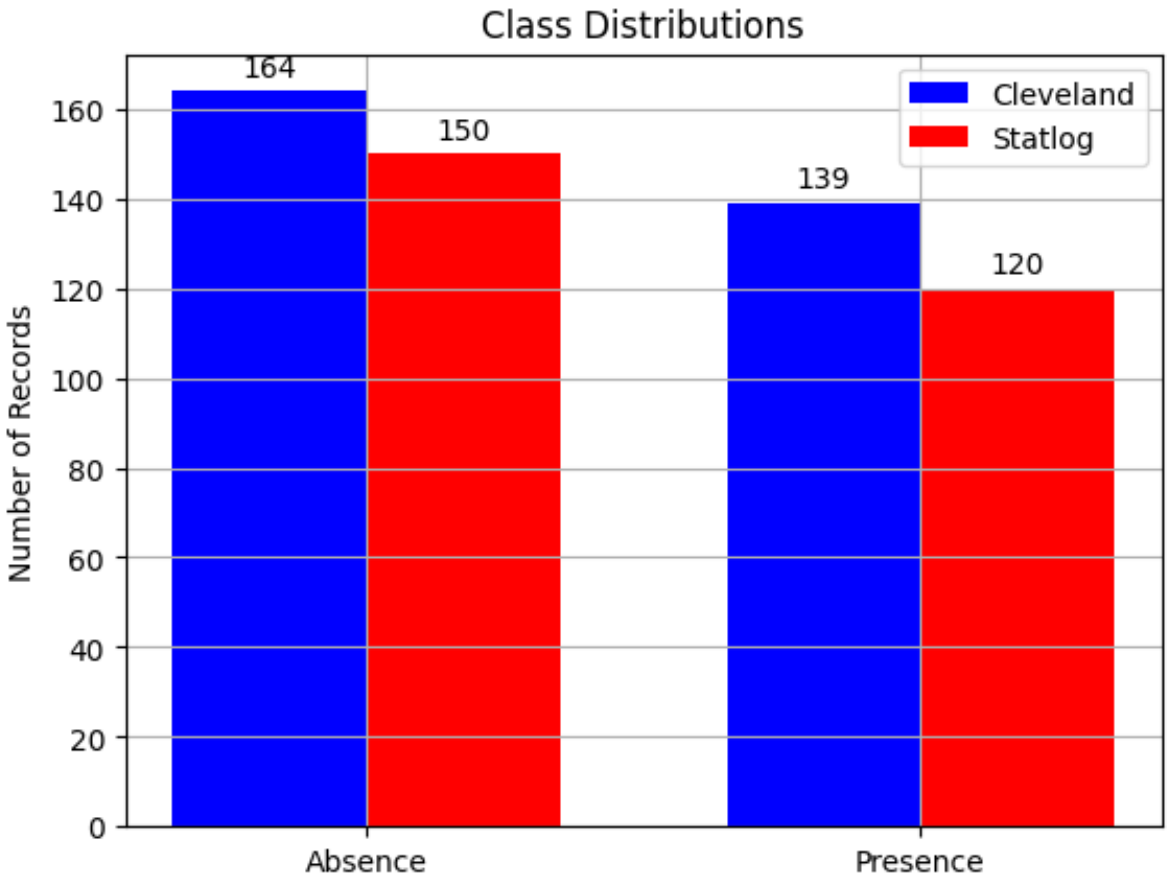
Datasets Description

Cleveland Dataset:

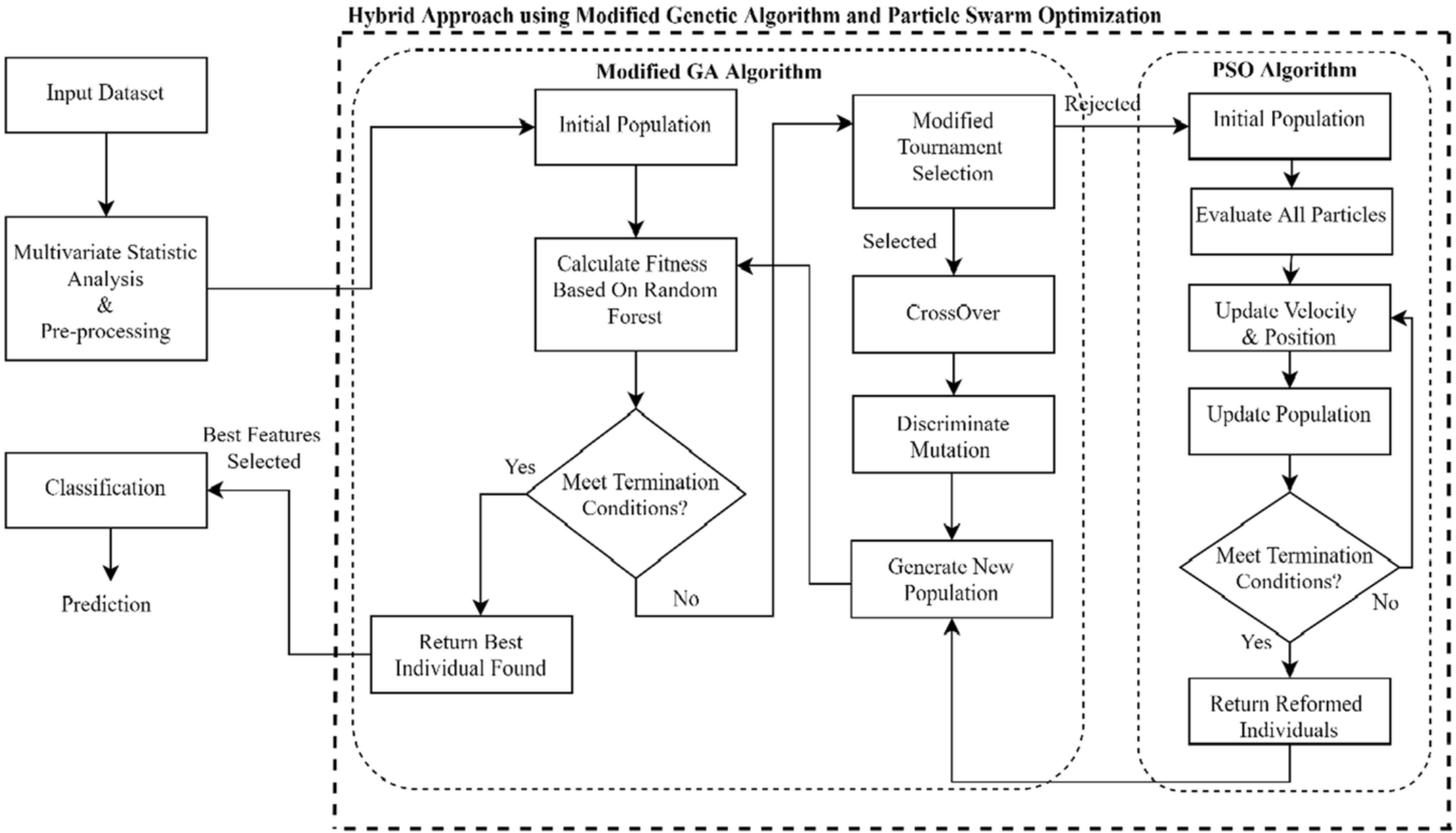
- 303 samples, 13 features
- Target: 0 = healthy, 1 = diseased

Statlog Dataset:

- 270 samples, same format
- Used for cross-validation and robustness check



Workflow of GAPSO:



Methodology Overview

1. Multivariate statistical analysis to rank features
2. GAPSO or GWO performs feature selection
3. Selected features passed to RF or XGB classifiers
4. Results evaluated using cross-validation

GAPSO Feature Selection

- Genetic Algorithm (GA): global search
- Particle Swarm Optimization (PSO): repairs non-fit solutions
- Combined = GAPSO
- Fitness function: classification accuracy using RF/XGB

```
 Cleveland Dataset Evaluation
GAPSO + Random Forest Accuracy: 0.8444 ± 0.0738
GAPSO + XGBoost Accuracy: 0.8013 ± 0.0826

 Statlog Dataset Evaluation
GAPSO + Random Forest Accuracy: 0.8333 ± 0.0799
GAPSO + XGBoost Accuracy: 0.7926 ± 0.0624
```

Dataset	Model	Best Accuracy	Selected Features
Cleveland	GAPSO + RF	0.857923	sex, cp, chol, fbs, restecg, oldpeak, slope, ca, thal
Cleveland	GAPSO + XGB	0.831475	age, sex, cp, trestbps, chol, fbs, restecg, exang, oldpeak, slope, ca, thal
Statlog	GAPSO + RF	0.866667	age, cp, fbs, restecg, exang, slope, ca, thal
Statlog	GAPSO + XGB	0.818519	age, sex, fbs, restecg, exang, oldpeak, slope, ca, thal

Grey Wolf Optimizer (GWO)

Grey Wolf Optimizer is a nature-inspired algorithm that mimics the leadership hierarchy and hunting behavior of grey wolves. It is used to select the most important features by simulating intelligent search patterns.

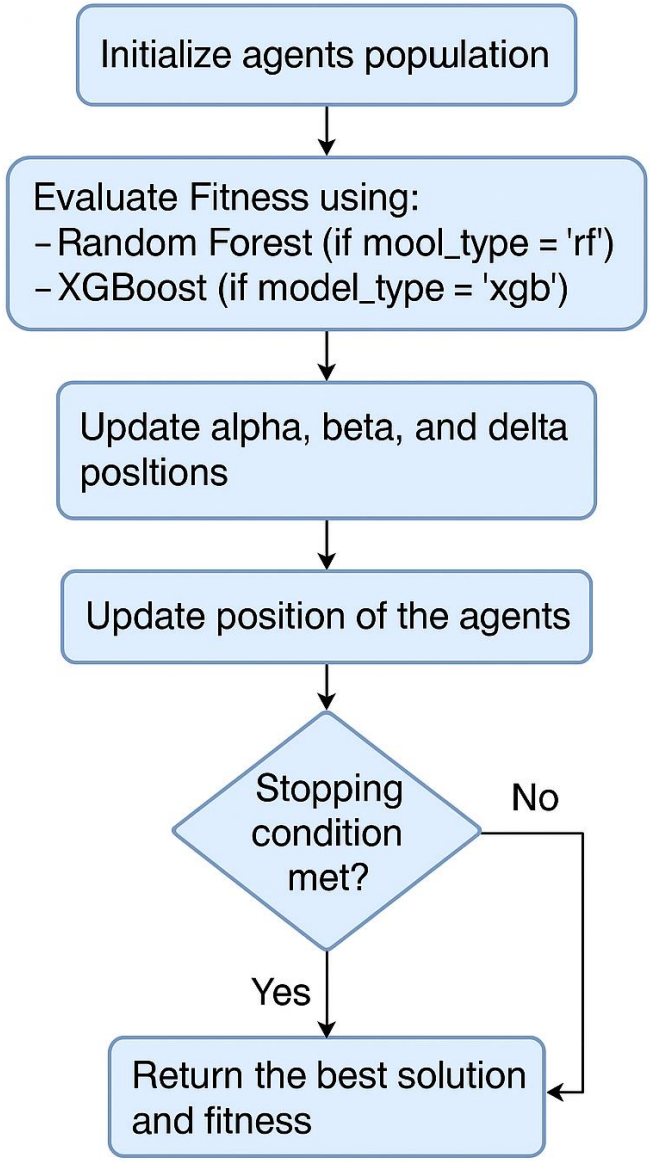
- We used GWO to select relevant features from the Cleveland and Statlog heart disease datasets.
- Each wolf (agent) represents a binary vector:
  1 = select feature, 0 = discard feature
- Wolves update their positions based on the top three leaders:
  Alpha (best), Beta (second), Delta (third).
- The fitness of each wolf is calculated using model accuracy from a classifier (Random Forest and XGB classifiers).

GWO Feature Selection - Cleveland Dataset

Model	Accuracy	Precision	Recall	F1 Score	AUC
GWO-RF	0.824624	0.831232	0.791209	0.804159	0.896769
GWO-XGB	0.831183	0.846564	0.792308	0.811689	0.889954

GWO Feature Selection - Statlog Dataset

Model	Accuracy	Precision	Recall	F1 Score	AUC
GWO-RF	0.848148	0.854346	0.816667	0.824828	0.916111
GWO-XGB	0.829630	0.843539	0.775000	0.800058	0.883889



Classifiers

Random Forest (RF)

An ensemble learning method that builds multiple decision trees and merges their predictions for higher accuracy.

XGBoost (XGB)

An advanced boosting algorithm optimized for speed and performance, often outperforming traditional models.

- **Model Integration:** RF and XGB were used with GAPSO and GWO feature selectors to create robust predictive hybrids

Classifier Performance Comparison on GAPSO

Cleveland Dataset

Model	Accuracy (Mean ± Std)	Precision (Mean ± Std)	Recall (Mean ± Std)	F1 Score (Mean ± Std)	AUC (Mean ± Std)
RandomForest	**0.8246 ± 0.0796**	0.8178 ± 0.0854	0.7786 ± 0.1214	0.8015 ± 0.0801	**0.9027 ± 0.0504**
DecisionTree	0.7286 ± 0.0965	0.6912 ± 0.0912	0.7214 ± 0.1334	0.7066 ± 0.0930	0.7186 ± 0.0981
NaiveBayes	0.8182 ± 0.0736	0.8173 ± 0.0852	0.7929 ± 0.1172	0.7991 ± 0.0815	0.8911 ± 0.0598
SVM	0.8082 ± 0.0735	**0.8312 ± 0.1072**	0.7566 ± 0.1153	0.7831 ± 0.0799	**0.9053 ± 0.0433**
XGBoost	0.7751 ± 0.0797	0.7736 ± 0.1085	0.7500 ± 0.1288	0.7526 ± 0.0857	0.8736 ± 0.0720

Statlog Dataset

Model	Accuracy (Mean ± Std)	Precision (Mean ± Std)	Recall (Mean ± Std)	F1 Score (Mean ± Std)	AUC (Mean ± Std)
RandomForest	**0.8370 ± 0.0554**	0.8579 ± 0.1185	0.7750 ± 0.1057	0.8000 ± 0.0502	0.8819 ± 0.0574
DecisionTree	0.7519 ± 0.0643	0.7619 ± 0.1413	0.7083 ± 0.0854	0.7196 ± 0.0524	0.7542 ± 0.0673
NaiveBayes	0.8259 ± 0.0575	0.8263 ± 0.0970	0.7917 ± 0.1070	0.8002 ± 0.0659	0.8878 ± 0.0637
SVM	**0.8407 ± 0.0551**	**0.8665 ± 0.0874**	0.7750 ± 0.1397	**0.8071 ± 0.0778**	**0.8867 ± 0.0580**
XGBoost	0.8000 ± 0.0848	0.8110 ± 0.1301	0.7667 ± 0.0972	0.7763 ± 0.0710	0.8678 ± 0.0491

Feature Reduction Summary (GAPSO vs. GWO)

Dataset	Total Features	GAPSO-RF	GWO-RF	GAPSO-XGB	GWO-XGB
Cleveland	13	9	7	12	10
Statlog	13	8	10	9	9

- **Both GAPSO** and **GWO** reduced dimensionality from the full **13 features**.
- **GWO-RF** in Cleveland achieved the best performance using only **7 features.**
- **GAPSO** often selected more features than **GWO**.
- Feature reduction improves model **efficiency, training speed, and clarity.**

```
Feature Dimension Summary
Cleveland (Total): 13 | GAPSO-RF: 9 | GWO-RF: 7 | GAPSO-XGB: 12 | GWO-XGB: 10
Statlog   (Total): 13 | GAPSO-RF: 8 | GWO-RF: 10 | GAPSO-XGB: 9 | GWO-XGB: 9
```

Accuracy Impact of Feature Selection (GAPSO vs. GWO)

Dataset	Model	Full Features	GAPSO-Selected	GWO-Selected
Cleveland	RF	0.8248	0.8246	**0.8345**
	XGB	0.8080	0.8013	**0.8312**
Statlog	RF	0.8259	0.8296	**0.8593**
	XGB	0.8111	0.7926	**0.8296**

- GWO achieved the highest accuracy across all models.
- Feature selection reduced dimensions without losing performance.
- GWO outperformed GAPSO and full models.
- Works effectively on both Cleveland and Statlog datasets.

```
 Statlog FS Evaluation
GAPSO-RF - All Features: 0.8259 | Selected: 0.8296
GWO-RF   - Selected Features: 0.8593
GAPSO-XGB - All Features: 0.8111 | Selected: 0.7926
GWO-XGB   - Selected Features: 0.8296

 Cleveland FS Evaluation ===
GAPSO-RF - All Features: 0.8248 | Selected: 0.8246
GWO-RF   - Selected Features: 0.8345
GAPSO-XGB - All Features: 0.8080 | Selected: 0.8013
GWO-XGB   - Selected Features: 0.8312
```

Conclusion

- Hybrid feature selection using GAPSO and GWO with RF and XGB.
- Tested on Cleveland and Statlog heart disease datasets.
- GWO outperformed GAPSO in both accuracy and feature reduction.
- Achieved high accuracy with fewer features, improving efficiency.

Thank You!